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Optimization of test and maintenance intervals based on risk and cost

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A general procedure is presented for optimizing the test and maintenance intervals of safety related systems and components. The method is based on minimizing the total plant-level cost under the constraint that the total accident frequency (risk) remains below a set criterion.

The measure of risk is the time-average value of an accident rate. The probabilities of component failures and other basic events are linear functions or inversely proportional to the test or maintenance intervals. Human errors and common cause failures are included in the formalism. Analytical results are obtained for single components and simple systems while a numerical procedure is given for obtaining optimal intervals for complex plants with multiple systems and initiating events.

1 INTRODUCTION

Engineered safety systems are usually standby systems that are tested periodically to reveal and repair failures that may have occurred since the previous activation or inspection. Downtime (unavailability) can be caused by testing or by failures and repairs caused by activations or occurring during standby, as well as by human errors associated with tests. Frequent testing increases the testing costs while infrequent testing leads to increasing downtime and risk. It is obvious that optimal test intervals usually exist for minimizing costs while satisfying the safety goal.

Early optimizations of single component test intervals were based on minimizing the time-average unavailability without cost considerations.[1–4] These models are now expanded in Section 2 to include cost minimization together with a risk or unavailability criterion. It is also possible that a single component is involved in two different periodic activities: testing and maintenance. Optimization of two periods is not a trivial exercise, as is illustrated by an example in Section 4.1.

For systems with multiple components it is not generally possible to solve optimal test intervals in analytical form, even if only the system unavailability is optimized without cost consideration. Although analytical unavailability equations are available for many m-out-of-n parallel redundancy systems (e.g. literature listed in Ref. 4), minimization requires a numerical solution with computer codes like ICARUS,[5] or trial and error by using time-dependent unavailability codes such as FRANTIC.[6]

Some studies on acceptable test intervals have set criteria for individual components such that the risk increase during standby unavailability should be a certain fraction of the average risk.[7] While this can be reasonable for risk-important components it does not consider costs or fully account for the fact that many components can belong to the same testing group. It can also lead to unrealistically long test intervals for a large number of less important components.

Test-caused failures have also been included in optimizing system test intervals on the basis of risk.[8,9] A general algorithm has been suggested for solving test intervals that minimize the system average unavailability (without cost consideration).[10]

Recently test interval calculation examples have been reported for more complex systems or plants using, as a criterion, an upper limit for the time-dependent risk (accident frequency).[11,12] Such calculations are believed to reduce the number of test episodes in some sense (at least compared to arbitrarily specified values), but they do not consider the fact that different testing groups have different numbers of components involved and require different amounts of resources.

It is also possible to study the risk impact of changing the value of a specific test interval and use the result as a basis for extending the interval.[13] If this is systematically applied to testing groups with high cost and low risk sensitivity, a near-optimal selection of intervals might be obtained. However, some or all intervals might need to be reduced to ensure that the risk remains acceptable. It is the objective of this paper to develop a systematic method to determine exactly which intervals to increase or decrease, and by how much, to minimize the total cost. The procedure is presented in Sections 3 and 4. Accident costs are included in the formalism in Section 5. The procedure is formulated so that it is easy to adopt in connection with fault tree codes that identify and manipulate terms (minimum cut sets) based on event identifications rather than keeping track of the time-dependencies of all terms explicitly. The procedure is illustrated with several examples, including common cause failures and human errors.

The essential features in this paper are (1) explicit minimization of cost instead of (or in addition to) minimizing the risk or unavailability, (2) considering the total plant-level cost and risk rather than individual components or systems, (3) inclusion of preventive maintenance intervals and costs in addition to surveillance tests or inspections, and (4) incorporating both standby and normally operating components, human errors and common cause failures.

The position taken in this paper is that the time-average risk is the most important safety criterion. Even if technical specifications of a plant may have action rules concerning simultaneous failures or allowed repair times, all these are included (or should be) in a correct risk model for the average accident rate.

2 OPTIMIZATION AT COMPONENT LEVEL

In this Section the optimization problem and solutions are formulated for a single component, or a train of components in series. A general unavailability model is first defined in Section 2.1, as a function of a test or maintenance interval T. Finding the value of T that minimizes the unavailability is explained in Section 2.2. The costs associated with testing (or maintenance) are included in the formalism and used as the optimization criterion in Section 2.3. Minimizing the total cost is the primary objective while a risk or unavailability limit sets an additional constraint for the optimization. It will be seen throughout this paper that minimizing the cost generally leads to longer characteristic intervals (while satisfying a risk limit) than minimizing the risk alone.

2.1 Component model

The time-average unavailability of a general standby component with test interval T can be presented as

$$u = u(T) = \rho + \frac{\Delta}{T} + z\lambda T \qquad (1)$$

with $z = \frac{1}{2}$ and parameters ρ, Δ and the failure rate λ independent of T. (We assume that $u \leq 0{\cdot}1$ for eqn (1) to hold, as is usually the case in practice.) Each of the parameters can be a function of other parameters due to different causes of unavailability. What is important for the optimization approach is that any unavailability contribution can be presented as a term that is either (1) independent of T, (2) proportional to T, or (3) inversely proportional to T.

Analyzing published detailed component models[4,6,14,15] and failure modes as to their dependency on time parameters indicates that ρ can include contributions (terms) such as

—probability of failure due to a demand (an initiating event),
—probability of a human error of omission (failure to return to service after a test),
—repair time unavailability contributions of any failures caused during standby,
—probability of failure during a mission time,
—unavailability due to monitored failures (detected immediately).

The outage time parameter Δ can consist of

—mean downtime due to a test or maintenance carried out with intervals T,
—repair time unavailability contributions of any failures caused by tests.

Equation (1) can also be applied directly for a train of several components in series when the whole train is tested simultaneously with intervals T. In such a case ρ and λ are the *sums* of the corresponding parameters of the individual components while Δ is the downtime per test or maintenance for the whole train.

One should notice that eqn (1) is valid also for normally operating (rather than standby) components. For those the constant term ρ accounts for normal unavailability due to failure repair times $\tau[\rho = \lambda\tau/(1 + \lambda\tau)]$ while Δ can be the mean outage time due to maintenance carried out with intervals T. The last term $z\lambda T$ can account, at least approximately, for a failure rate increase between maintenances when z is properly defined. (A method to handle more general time-dependent failure rates is outlined in Section 4.1.)

2.2 Unavailability optimization

Assuming that risk optimization is equivalent to minimizing the time-average unavailability u, the

optimal test interval can be solved by setting the derivative du/dT of eqn (1) equal to zero. This yields

$$\hat{T} = \sqrt{\frac{2\Delta}{\lambda}}. \tag{2}$$

However, one should not blindly use this result as the only criterion for selecting T. With $T = \hat{T}$ the last two terms of eqn (1) are equal, and the sum is rather insensitive to T within a factor of 2 from $\hat{T}$. On this basis alone one is entitled to test less frequently than exactly at intervals $\hat{T}$ without increasing the risk unreasonably. There is another important consideration. In case of $\Delta = 0$ (or small Δ), eqn (2) yields small $\hat{T}$ (i.e. frequent testing) even when $\frac{1}{2}\lambda\hat{T}$ could be significantly smaller than ρ. In relative terms there is no benefit in selecting $\frac{1}{2}\lambda T$ orders of magnitude smaller than ρ. It should be quite acceptable to select the value of T anywhere in the interval

$$\hat{T}/2 < T < 2\hat{T} + \rho/\lambda. \tag{3}$$

For example, with $\rho = 10^{-3}$, $\lambda = 10^{-5}$/hr, $\Delta \times 0{\cdot}2$ hr, eqn (2) yields $\hat{T} = 200$ hr. The unavailability at $T = \hat{T}$ is $u = \rho + \lambda\hat{T} = 3 \times 10^{-3}$. At $T = 2\hat{T} + \rho/\lambda = 500$ hr the unavailability is $u = 3{\cdot}9 \times 10^{-3}$, only 30% higher than the absolute minimum. This is a small difference compared to the uncertainty bounds usually assessed for reliability parameters.

2.3 Cost-based optimization

There are always some costs associated with testing and maintenance. Selecting an interval that minimizes the unavailability can lead to an expensive maintenance program not warranted by risk considerations. In this section a cost function is defined and minimized, leading to less frequent testing while the unavailability still remains below a set limit.

Assume that the average cost associated with each periodic test (or maintenance) is C. The total cost over a long time t is Ct/T. It is appropriate to use the cost rate $y(T) = C/T$ as a function to be minimized.

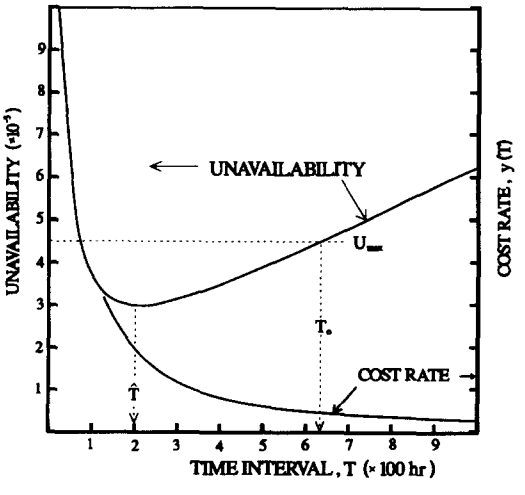


Fig. 1. Unavailability and cost rate as functions of T.

Assume that a maximum unavailability limit u_{max} has been specified by authorities or selected based on the results of a risk assessment (e.g. using importance measures such as Risk Achievement).[7–9] Because $y(T)$ is a decreasing function of T, one wishes to find T as large as possible within the limits of allowable risk. The optimal solution $T = T_o$ is now the largest real solution (if such exists) of the equation

$$u(T) \leq u_{max}, \tag{4}$$

where the l.h.s. is according to eqn (1) (with $z = \frac{1}{2}$). The situation is illustrated in Fig. 1 where both $y(T)$ and $u(T)$ are presented as functions of T. It shows that $u(T)$ is rather flat around $\hat{T}$. It is obvious that eqn (4) does not have a real solution if even the absolute minimum of eqn (1) is larger than u_{max}, i.e. $u_{max} < \rho + \lambda\hat{T}$. If $u_{max} \geq \rho + \sqrt{2\lambda\Delta}$, there are solutions for eqn (4), and the largest one is obtained with exact equality $u = u_{max}$:

$$T_o = \lambda^{-1}\{u_{max} - \rho + [(u_{max} - \rho)^2 - 2\lambda\Delta]^{1/2}\}. \tag{5}$$

With the numerical values of Δ, ρ and λ in the example of the previous section, eqn (4) has no real solution if $u_{max} < 3 \times 10^{-3}$. With $u_{max} = 10^{-2}$, eqn (5) yields $T_o = 1777{\cdot}5$ hr, significantly larger than $\hat{T} = 200$ hr. The amount of testing work is reduced by an order of magnitude by using T_o instead of $\hat{T}$.

It is interesting to notice the following characteristics of the solution (eqn (5)): (a) $T_o \geq \hat{T}$, (b) T_o does not depend on u_{max} and ρ individually but only through the difference $u_{max} - \rho$, and (c) T_o is independent of C. Thus, the result is insensitive to any uncertainties in the cost estimate.

3 PLANT-LEVEL OPTIMIZATION

In this section, the cost and risk functions are defined, and the derivatives of both are developed in terms that are suitable for the optimization procedure. The main definitions and assumptions are the following:

(1) there are $I \geq 1$ characteristic time periods T_i, $i = 1, 2, \ldots, I$ at the plant or system under study. Each T_i is an interval of periodic tests or maintenance actions for a certain group of components, a *timing group*. The components of a timing group are tested or maintained with the same interval, but not necessarily simultaneously. Thus, components of a timing group can belong to the same train or to different trains of a redundant system;

(2) any component can be a member of one or more timing groups, e.g. one group (and period) for testing and another group for maintenance;

(3) the unavailability-state of each component is

modeled by one or more *basic events* in the logic plant risk model. Different basic events of one component can be associated with different timing groups, e.g. some with testing, some with maintenance;

(4) multiple component unavailabilities due to repeated human errors or common-cause failures are also modeled as basic events in the plant (system) logic model;
(5) the basic events are mutually independent;
(6) the risk (accident rate) function F can be presented as a linear function of the initiating event frequencies and the basic event probabilities (i.e. in the usual 'sum of products' form);
(7) the basic event probability (b.e.p.) u_k of each of the K basic events depends on no more than one T_i. Each u_k $(k = 1, 2, \ldots, K)$ can be independent of any T_i or proportional $\sim T_i$ or $\sim T_i^{-1}$ (according to the general component model, eqn (1)).

It will be seen that these assumptions cause no severe limitations to current risk assessment practice. (In fact, assumption 7 can be relaxed by permitting u_k to be a sum of products of independent factors that each depend on some T_i. This will be demonstrated in Section 4.1.)

The times of interest, T_i, are variables subject to optimization. If any test or maintenance interval is a fixed constant, predetermined by rules or criteria other than optimization, it is not included in the set of I variables as defined. The unavailabilities of the basic events associated with such fixed intervals are constants in the current problem, independent of any of the periods T_i, $i = 1, 2, \ldots, I$. Vector notation $\mathbf{T} = \{T_i\}$ is used to indicate any function of all T_i's.

3.1 Cost function

At the plant level the cost rate function to be optimized (minimized) is

$$y(\mathbf{T}) = \sum_{i=1}^{I} \frac{C_i}{T_i}, \tag{6}$$

a sum of terms of which each is inversely proportional to one T_i. Each C_i is a measure of cost per action, carried out with intervals T_i.

There are several alternatives for selecting the cost constants C_i. With $C_i = Y$, the period of plant operation (e.g. $Y = 1$ yr), one is minimizing the total number of group tests or maintenances in time Y. With $C_i = Yn_i$, where n_i is the number of component trains belonging to timing group i, one is minimizing the total number of test or maintenance episodes in time Y. Finally, selecting $C_i = Yn_ic_i$, where c_i is a measure of cost or workload per one train episode in timing group i, one is minimizing the total cost in time Y.

The form of y is such that one hopes to maximize every T_i to the extent allowed by a risk constraint. The problem in general terms is to solve the times T_i that minimize the cost function

$$y = \sum_{i=1}^{I} \frac{C_i}{T_i} = \text{min!, under the condition } F(\mathbf{T}) \le R, \tag{7}$$

i.e. under the constraint that the risk function $F(\mathbf{T})$ does not exceed a set limit value R. F will be defined more specifically in Section 3.2. It is also assumed that non-negative upper and lower limits Y_i^+ and Y_i^- may be given:

$$Y_i^- \le T_i \le Y_i^+, \quad i = 1, 2, \ldots, I. \tag{8}$$

This is called the admissible region of $\mathbf{T}$. Some or all of the limits can be 0 or ∞. Equation (1) indicates that in practice Y_i^- should be larger than any Δ of components associated with timing group $i(T_i)$, and Y_i^+ should be smaller than any λ^{-1} of those components.

Some basic features of the problem are as follows:

Case A: If $F(\mathbf{Y}^+) \le R$, then the optimal solution is $\mathbf{T} = \mathbf{Y}^+$, because y cannot be made any smaller than the absolute minimum cost $y_o = y(\mathbf{Y}^+)$.

Case B: If $F(\mathbf{Y}^+) > R$, the optimal solution $\mathbf{T} = \mathbf{T}_o$, if one exists, satisfies $F(\mathbf{T}_o) = R$ (i.e. exact equality holds rather than $F < R$).

Proof: If there is a solution $\mathbf{T}$ such that $F(\mathbf{T}) < R$, it is always possible to increase any T_i (thereby reducing cost) until $T_i = Y_i^+$ or until $F(\mathbf{T}) = R$. Thus, at the end either $F(\mathbf{T}_o) = R$, or all $T_i = Y_i^+$.

Thus, in the case where $F(\mathbf{Y}^+) > R$, the problem can be reformulated

$$y = \sum_{i=1}^{I} \frac{C_i}{T_i} = \text{min!, under the condition } F(\mathbf{T}) = R. \tag{9}$$

The problem is qualitatively similar to Fig. 1 when F (in place of 'unavailability') is considered as a function of any single T_i, F has a limit R (in place of u_{max}) and the cost rate is a decreasing function of T_i. However, multiple intervals cannot be solved as simply as a single interval in Section 2. Of cause, this problem does not have a solution if the minimum of F is larger than R.

3.2 Risk function

The risk function $F(\mathbf{T})$ represents the frequency of accidents with specified consequences, e.g. the core damage frequency of a nuclear power plant. The Boolean logic form of an accident can be presented as

$$A = \sum_{\nu} \phi_\nu \cdot \sum_{\mu} M_{\nu\mu} \tag{10}$$

where Φ_ν indicates the occurrence of an initiating event of type ν, and $M_{\nu\mu}$ is minimum cut set (MCS) μ

relevant to event v, i.e. a set of unavailability states (basic events) of safety components. Each $M_{v\mu}$ is a logic product (intersection) of individual basic event Boolean variables Z_k ($k = 1, 2, \ldots, K$).

The algebraic probability of A per unit time is the accident frequency, $F(\mathbf{T})$. It can be presented in the form

$$F = \sum_v f_v G_v, \tag{11}$$

where f_v is the frequency of initiating event type v, and the probability G_v is a sum of *terms*, products of basic event probabilities $u_k = P(Z_k)$, relevant to initiating event type v. Symbolically,

$$G_v = \sum_\delta d_\delta u_k \cdots u_r \tag{12}$$

where $d_\delta = \pm 1$. In rare-event approximation all $d_\delta = 1$ and each term (product) is a probability of an MCS. In a complete exact expression there can be negative terms ($d_\delta = -1$) due to intersections of MCS's according to the Poincare's theorem. Most important for the solution procedure is that *F is linear in terms of any basic event probability u_k*. We assume here that common cause failures (CCF) as well as human errors (HE) and component failure modes are all modeled as basic events in the system logic models.

According to the general component model eqn (1), each b.e.p. u_k can be independent of any T_i or proportional $\sim T_i$ or $\sim T_i^{-1}$. The same is true for CCF's and HE's, as will be shown. Thus, every term (eqn (12)) is proportional to some powers of some T_i's, and some of the terms might be independent of any T_i. It is permissible but not necessary to delete from F all terms that are independent of all T_i's if one subtracts the same amount from R.

From the linearity of F in terms of any u_k it follows that

$$\frac{\partial F}{\partial u_k} = \frac{\sum \text{of terms containing } u_k}{u_k}. \tag{13}$$

This is always positive in a coherent plant model in which increasing failure probability always increases risk.

This fact has a Corollary: If there is no basic event proportional to a particular T_j, then F is a nonincreasing function of that T_j. One can select the optimal value $T_j = Y_j^+$, and eliminate this component of $\mathbf{T}$ from further consideration.

Assume from now on that such T_j's have been eliminated from the problem and are not among the times to be optimized. The optimization problem is then (unless $F(\mathbf{Y}^+) \le R$)

$$y = \sum_{i=1}^{I} \frac{C_i}{T_i} = \min!, \text{ under the condition } F(\mathbf{T}) = R, \tag{14}$$

and for every T_i there is at least one basic event with $u_k \sim T_i$. One should notice that the absolute values of C_i do not influence the optimization, only the relative values (i.e. y can be multiplied by any constant >0).

The optimization procedure to be described is relevant to cases $I \ge 2$. In case $I = 1$ the condition $F = R$ uniquely determines the single time interval T_1 if such exists [or $T_1 = Y_1^+$ if $F(Y_1^+) \le R$].

3.2.1 Derivatives of $F(\mathbf{T})$

Definitions:

K_i^+ = subset of basic event indices k ($1 \le k \le K$) with probability proportional to T_i, i.e. $u_k = z_k \lambda_k T_i$ with some constant $z_k > 0$ and failure rate $\lambda_k > 0$;

K_i^- = subset of basic event indices k' ($1 \le k' \le K$) with probability inversely proportional to T_i, i.e. $u_{k'} = \Delta_{k'}/T_i$ with some constant $\Delta_{k'} > 0$.

Thus, $K_i = K_i^+ \cup K_i^-$ is the set of basic events (indices) that belong to timing group i.

The derivative of F with respect to a specific T_i is

$$\frac{\partial F}{\partial T_i} = \sum_{k \in K_i} \frac{\partial F}{\partial u_k} \frac{\partial u_k}{\partial T_i}, \tag{15}$$

where the sum is over all terms of F containing at least one basic event with $u_k = z_k \lambda_k T_i$ ($k \in K_i^+$) or $u_{k'} = \Delta_{k'}/T_i$ ($k' \in K_i^-$). From these one obtains

$$\frac{\partial u_k}{\partial T_i} = \frac{u_k}{T_i} \quad \text{for} \quad k \in K_i^+, \tag{16}$$

$$\frac{\partial u_{k'}}{\partial T_i} = -\frac{u_{k'}}{T_i} \quad \text{for } k' \in K_i^-. \tag{17}$$

Defining

s_k^+ = sum of terms in F containing a specific

$$u_k = z_k \lambda_k T_k, \quad (k \in K_i^+), \tag{18}$$

$s_{k'}^-$ = sum of terms in F containing a specific

$$u_{k'} = \frac{\Delta_{k'}}{T_i}, \quad (k' \in K_i^-), \tag{19}$$

and taking into account eqns (13), (15), (16) and (17) one obtains

$$\begin{aligned} \frac{\partial F}{\partial T_i} &= \frac{S_i}{T_i}, \quad S_i = S_i^+ - S_i^-, \\ S_i^+ &= \sum_{k \in K_i^+} s_k^+ \\ S_i^- &= \sum_{k' \in K_i^-} s_{k'}^-. \end{aligned} \tag{20}$$

It is quite common that many terms that include $u_{k'} = \Delta_{k'}/T_i$ also contain some $u_k = z_k \lambda_k T_i$, canceling terms out from S_i. One should notice that S_i^+ can have the same term more than once because several basic event probabilities in one term can be proportional to

T_i. On the other hand, S_i^- normally does not have repeated terms because there is a rule not to test or maintain redundant trains simultaneously (to avoid making multiple components unavailable at the same time). Based on this same reason S_i^+ is normally a polynomial function of T_i of degree ≥ 0, while S_i^- can also be such a polynomial but additionally have terms $\sim T_i^{-1}$. Consequently, S_i can be negative for small values of T_i.

Considering an unconstrained optimization problem without any limits for the time variables T_i, the problem defined by eqn (14) can be reformulated as

$$\sum_{i=1}^{I} \frac{C_i}{T_i} + L[F(\mathbf{T}) - R] = \min!, \qquad (21)$$

where L is a Lagrange multiplier (constant). Setting the derivatives of eqn (21) equal to zero yields the conditions

$$-\frac{C_i}{T_i} + LS_i(\mathbf{T}) = 0, \quad i = 1, 2, \ldots, I. \qquad (22)$$

This shows that at the optimal point $\mathbf{T}_o$ the ratios $C_i/(T_iS_i)$ are equal and L is equal to this cost/risk sensitivity ratio. However, finite limits $Y_i^- \leq T_i \leq Y_i^+$ can complicate the problem and make eqn (22) invalid at the point where eqn (14) is satisfied in the admissible region. The general optimization procedure will be presented in Section 4. (The idea of first solving the problem without limits for T_i, and then setting those limits *afterwards* to Y_i^+ or Y_i^- does not generally yield the optimum solution.)

3.2.2 Coefficients z_k and common cause failures

For a single component (or train) it has been pointed out (eqn (1)) that the coefficient $z_k = \frac{1}{2}$ gives the correct time-average unavailability term for basic events with probability $\sim T_i$ ($k \in K_i^+$). It has also been shown that different values of z_k are needed for basic events in parallel redundancy systems, and the values depend on the success criterion and on how the tests are staggered.[16] The same is true for common cause failures. In a common cause basic event probability, $u_k = z_k\lambda_k T_i$, λ_k is the rate of failure of a *specific group of multiple components*. Table 5 of Ref. 16 gives the factors z_k between 0·125 and 0·669, for m-out-of-n:G systems ($1 \leq m \leq n \leq 4$) for three testing schemes: (1) sequential testing, (2) staggered testing, and (3) staggered testing with extra testing of other trains whenever a failure is observed in one train. Values are given in Ref. 16 for single failures as well as for common cause failures of any multiplicity.

3.2.3 Human errors

It was pointed out in Section 2.1 that the probability of a human error of omission (failure to return a component to service after a test) can be taken into account as a constant term (as ρ in eqn (1)) for a single component. This is usually satisfactory also for components in redundancy systems with staggered testing schemes because the dependency between tests of different trains is small. However, especially with sequential testing, there is a higher probability to repeat an error in other trains if one is made in the first test of a testing cycle. If the probability of the first error is γ_0 and γ_n is the conditional probability of $(n+1)^{\text{th}}$ error, given n preceding errors, then a group of n components becomes unavailable with probability $\rho = \gamma_0, \gamma_1, \ldots, \gamma_{n-1}$ due to human errors.[17] This is usually larger than $\rho' = \gamma_0^n$ that results when only independent errors are modeled. Thus, the difference has to be modeled as a common cause event impacting a group of n trains to avoid underestimating the risk. Essential for the optimization procedure here is that the human error multi-train unavailabilities, e.g. $\rho - \rho'$, do not depend on the value of T_i. This is because a set of errors, if it occurs, remains in the system for the whole period T_i (or a certain fraction of T_i). Similarly, operator errors made during a transient (e.g. failure to operate a safety system) are typically system level basic events impacting multiple components, and independent of any T_i.

Even if derivations in eqn (15) need not be done with respect to constant u_k's, constant unavailability events are normally present in the terms added up in eqns (18)–(20). Thus, it is necessary to model repeated human errors properly.[18]

3.2.4 Modeling example

To illustrate the concepts introduced so far, consider a parallel 1-out-of-2:G system of identical components, tested sequentially. The unavailability of this system can be modeled by $K = 8$ basic events $Z_1, \ldots, Z_8$ with probabilities (unavailabilities) $u_1 = u_2 = \gamma_0$, $u_3 = u_4 = 1/\sqrt{3}\,\lambda_{1/2}T$, $u_5 = u_6 = \Delta/T$, $u_7 = \gamma_0\gamma_1 - \gamma_0^2$, $u_8 = \frac{1}{2}\lambda_{2/2}T$, respectively. Thus, $K^+ = \{3, 4, 8\}$, $K^- = \{5, 6\}$. ($\lambda_{k/n}$ is the rate of failures of specific k components out of n components.)

The Boolean risk function is

$$A = \Phi\,.\,(Z_1Z_2 + Z_1Z_4 + Z_1Z_6 + Z_3Z_2 + Z_3Z_4 + Z_3Z_6 + Z_5Z_2 + Z_5Z_4 + Z_7 + Z_8).$$

(Notice that Z_5Z_6 is missing because simultaneous maintenance is not done.) In rare-event approximation F is obtained from A by replacing Φ with the initiating event frequency f and each Z_k by u_k. The sums of terms needed in the derivative equations are

$$s_3^+ = fu_3u_2 + fu_3u_4 + fu_3u_6,$$

$$s_4^+ = fu_1u_4 + fu_3u_4 + fu_5u_4,$$

$$s_8^+ = fu_8, \quad s_5^- = fu_5u_2 + fu_5u_4$$

and

$$s_6^- = fu_1u_6 + fu_3u_6.$$

Finally, the derivative is

$$\frac{dF}{dT} = \frac{S}{T}$$

with

$$S = f(u_3u_2 + u_3u_4 + u_1u_4 + u_3u_4 + u_8 - u_5u_2 - u_1u_6).$$

For this simple system the result can be verified directly.

With staggered testing the only differences are that Z_7 can be deleted ($\gamma_1 = \gamma_0, u_7 = 0$), $u_3 = u_4 = \sqrt{\frac{5}{24}}\lambda_{1/2}T$ and $u_8 = \lambda_{2/2}T/4$. Thus, the risk is lower with staggered testing than with consecutive testing.

The method may seem a little cumbersome in this simple problem. However, computer codes that solve large fault trees for total plant risk models usually manipulate minimum cut sets and basic events rather than cut set term dependencies on T_i directly. The method is easy to adopt for use in connection with such fault tree codes. Events belonging to different timing groups (K_i^+ and K_i^- for $i = 1, \ldots, I$) need to be identified so that a code can list and add up terms that contain specified events.

3.3 Multiple initiating events

A few simple examples with unconstrained times are analyzed in this section to demonstrate the method.

Consider a risk function of two time variables T_1 and T_2 that are associated with different systems and initiating events. For example, T_1 can be a test interval for an emergency core cooling system (ECCS) that responds to loss of coolant accidents (LOCA) while T_2 is a test interval for an emergency power supply system (EPSS) responding to loss of offsite power (LOSP) events. In the simplest case the risk function can be presented as

$$F = aT_1^n + bT_2^m. \tag{23}$$

Here a and b can depend on other test and maintenance intervals as well as initiating event frequences but can be considered constants in this optimization problem. (Also terms that are independent of both T_1 and T_2 have been subtracted from F before optimization.) The integers $n \geq 1$, $m \geq 1$ depend on whether common cause failures or other multiple failures dominate the risk. In this case optimization of the cost function $y = C_1/T_1 + C_2T_2$ by eqn (22) leads to optimal intervals

$$T_1 = \left(\frac{C_1}{nLa}\right)^{1/n+1}, \quad T_2 = \left(\frac{C_2}{mLb}\right)^{1/m+1}, \tag{24}$$

where L has to satisfy the equation

$$R = a\left(\frac{C_1}{nLa}\right)^{n/n+1} + b\left(\frac{C_2}{mLb}\right)^{m/m+1}. \tag{25}$$

In case of a general linear risk function

$$F = \sum_{i=1}^{I} a_iT_i \tag{26}$$

the optimal times $T_i = T_{i,0}$ are proportional to R according to

$$a_iT_{i,0} = R\sqrt{a_iC_i}\Big/\sum_{i=1}^{I}\sqrt{a_iC_i}. \tag{27}$$

In the case $I = 2$ this yields

$$T_{1,0} = \frac{R}{a_1(1+x)}, \quad T_{2,0} = \frac{Rx}{a_2(1+x)} \tag{28}$$

where

$$x = \left(\frac{a_2C_2}{a_1C_1}\right)^{1/2}.$$

Example: At a certain plant the current test intervals for both ECCS and EPSS are four weeks, and the acceptable risk is $F = R = 10^{-5}$/yr. The risk due to LOCA is 90% leaving 10% for LOSP. The unavailability of both systems is dominated by common cause failures, which justifies a linear risk model (for dominating terms).

This means that $R/a_1 = 746{\cdot}7$ hr and $R/a_2 = 6720$ hr. The test related costs of EPSS are nine times as high as those for ECCS (due to related fuel and maintenance), i.e. $C_2/C_1 = 9$. Equations (28) yield new optimal test intervals $T_1 = 373$ hr and $T_2 = 3360$ hr, without increasing risk. The cost reduction due to changing from four week intervals to the new values is 64%.

It is also possible that a dominating family of initiating events is protected against by two redundant systems, with different test intervals, T_1 and T_2. The risk function $F = aT_1T_2$ leads to optimum test intervals

$$T_1 = \left(\frac{RC_1}{aC_2}\right)^{1/2} \quad \text{and} \quad T_2 = \left(\frac{RC_2}{aC_1}\right)^{1/2}.$$

In all these examples the intervals increase with an increasing risk limit and depend on the cost ratios rather than the absolute values of C_1 and C_2. This makes the optimization rather insensitive to systematic errors in the cost estimates.

In all cases it is rather straightforward to evaluate how sensitive the risks and the costs are to possible deviations of T_i from the optimum values. Doubling the intervals (or R) with the risk eqn (26) would double the risk and reduce the cost by 50%. In case of $F = aT_1T_2$ the risk is quadrupled by doubling both T_1 and T_2.

4 OPTIMIZATION PROCEDURE

The starting point of general optimization is eqn (14) (because the case $F < R$ is relevant only when

$F(\mathbf{Y}^+) < R$, in which case $\mathbf{T} = \mathbf{Y}^+$ is the solution). An iterative procedure can be obtained by considering the risk and cost differentials from eqns (20) and (6) respectively:

$$dF = \sum_{i=1}^{I} \frac{\partial F}{\partial T_i} dT_i = \sum_{i=1}^{I} S_i(\mathbf{T}) \frac{dT_i}{T_i}, \tag{29}$$

$$dy = -\sum_{i=1}^{I} \frac{C_i}{T_i} \frac{dT_i}{T_i}. \tag{30}$$

If the condition $F = R$ is not satisfied with the initial values of $\mathbf{T}$, the first task is to use eqn (29) to make effective changes ΔT_i until the level $F = R$ is reached. After that one makes combinations of changes so that $\Delta F = 0$ while the cost is reduced, i.e. $\Delta y < 0$. In this part the idea is to increase T_i for i such that the *partial cost/risk-sensitivity ratio*

$$r_i = \frac{C_i}{T_i S_i(\mathbf{T})} \tag{31}$$

is large compared to the *total cost/risk sensitivity ratio.*

$$r = \sum_{i=1}^{I} \frac{C_i}{T_i} \Big/ \sum_{i=1}^{I} S_i(\mathbf{T}), \tag{32}$$

and *reduce* T_j if $r_j < r$ in such a way that F remains constant, i.e.

$$S_j \frac{\Delta T_j}{T_j} = -S_i \frac{\Delta T_i}{T_i} \tag{33}$$

(at least two values are changed at a time).

A step by step procedure is presented in Appendix 1. The total risk function F is assumed to be available, typically stored in a computer file in the form of terms (sum of products) and associated basic events. Usually F is known numerically at least in one admissible point with initial values $\mathbf{T}$.

4.1 A maintenance example

To illustrate the solution procedure consider a simple risk function

$$F = f\left(\tfrac{1}{2}\lambda T_1 + \frac{\Delta}{T_2}\right), \tag{34}$$

where

f = the initiating event rate (demand rate),
T_1 = interval of periodic tests or inspections,
T_2 = interval of periodic maintenance,
Δ = average duration of a maintenance outage,
λ = standby failure rate of a safety system.

(One can assume that any terms that are independent of both T_1 and T_2 have been deleted from F and subtracted from the risk criterion R.)

The cost rate function to be minimized is

$$y = \frac{C_1}{T_1} + \frac{C_2}{T_2}, \tag{35}$$

where C_1 and C_2 are the costs associated with testing and maintenance, respectively ($I = 2$).

The form of F is such that T_2 can be increased without limit because both risk and cost can be reduced by increasing T_2. A selected upper limit Y_2^+ for T_2 together with a condition $F = R$ dictates a unique value for T_1 without any further use of the cost function. Minimizing the risk alone would lead to $T_2 = Y_2^+$ and $T_1 = Y_1^-$.

A more genuine optimization problem can be formulated by taking into account that the failure rate λ increases in time until a maintenance is performed. Assuming a linear growth rate 2ε for λ, the average failure rate over a maintenance interval is

$$\lambda = \lambda_0(1 + \varepsilon T_2), \tag{36}$$

where λ_0 is the failure rate immediately after maintenance. The risk function now becomes

$$F = fu_1 + fu_1u_2 + fu_3 \tag{37}$$

as a sum of products, where $u_1 = \frac{1}{2}\lambda_0 T_1$, $u_2 = \varepsilon T_2$ and $u_3 = \Delta/T_2$. With numerical values $f = 0{\cdot}01$/yr, $\varepsilon = 10^{-4}$/hr, $\Delta = 20$ hr, $R = 10^{-4}$/yr, $C_1 = 1000$ \$ and $C_2 = 4000$ \$, the solution is presented in Appendix 2 following the procedure presented in Appendix 1, step by step. (The units of the variables are given in the early steps of the procedure.) The result is $T_1 = 785$ hr, $T_2 = 10\,734$ hr, and the minimum cost is $y = 1{\cdot}646$ \$/hr. (The same T_1 and T_2 are obtained whenever $C_2/C_1 = 4$, independent of the absolute cost values.)

It is interesting to notice that in this case u_2 is not really a probability but a factor that can have values larger than unity. The solution process works because the problem is mathematically in a proper form.

The risk (eqn (37)) as a function of T_2 is qualitatively similar to the operational unavailability obtainable by other models (e.g., Fig. 3 of Ref. 19). However, the methodology is not limited to the linear form of eqn (36), used here only as an example. It is possible to generalize to a polynomial of any degree q (or a Taylor series) as an ageing model between maintenance events,

$$\lambda = \lambda_0\left(1 + \sum_{i=1}^{q} \epsilon_i t^i\right), \quad 0 \le t < T_2.$$

The first part of the risk function, $f\frac{1}{2}\bar{\lambda}T$, where $\bar{\lambda}$ is the average of λ over T_2 can be written as

$$fu_0 + fu_0u_1 + fu_0u_1u_2 + \cdots + fu_0u_1 \cdots u_q,$$

where

$$u_0 = \tfrac{1}{2}\lambda_0 T_1, \quad u_1 = \tfrac{1}{2}\epsilon_1 T_2, \quad u_\nu = \frac{\nu\epsilon_\nu}{(\nu+1)\epsilon_{\nu-1}} T_2$$

for $v = 2, \ldots, q$. Thus, the risk function is in the form of a sum of products of factors that are linear in T_1 and T_2. This is directly suitable for the optimization procedure.

4.2 Existence and uniqueness of a solution

It is quite possible, of course, that the minimum of F is larger than R, in which case there is no solution. It is of interest to know that the minimum of F that may be reached (in step 5 of the optimization procedure, Appendix 1) is a unique global minimum. This can be proved, at least under the following *assumption*:

> F is presented as a rare event approximation i.e. as a sum of non-negative terms, each consisting of an initiating event frequency multiplied by a minimum cutset probability (product of a set of unavailabilities u_k).

Consider F as a function of any one particular T_i. F consists of three mutually exclusive sets of terms:

G_i^+ = set of terms in which the number of basic events $k \in K_i^+$ is larger than the number of basic events $k' \in K_i^-$;

G_i^o = set of terms in which the number of basic events $k \in K_i^-$ is equal to the number of basic events $k \in K_i^-$;

G_i^- = set of terms in which the number of basic events $k \in K_i^+$ is smaller than the number of basic events $k' \in K_i^-$.

Thus, every term in G_i^+ is proportional to $\sim T_i^n$ with an integer $n \geq 1$, every term G_i^o is independent of T_i, and every term in G_i^- is proportional to $\sim T_i^{-m}$, with an integer $m \geq 1$. Consider now the definitions of S_i^+, S_i^- and S_i (eqn (20)). Every term of G_i^o that is in S_i^- is equally repeated in S_i^+ so that the difference $S_i = S_i^+ - S_i^-$ does not contain any terms of G_i^o (terms independent of T_i). Both S_i^+ and S_i^- can contain terms from both sets G_i^+ and G_i^-, but in the difference, S_i, equal terms cancel out, leaving positive terms $\sim T_i^n$ and negative terms $\sim T_i^{-m}$. Thus, S_i can be written as

$$S_i = P_i^+(T_i) - P_i^-\left(\frac{1}{T_i}\right), \tag{38}$$

where P_i^+ and P_i^- are polynomials of degree ≥ 1 with non-negative coefficients and $P_i^+(o) = P_i^-(o) = 0$ ($\lim P_i^- = 0$ for $T_i \to \infty$).

If $P_i^- \equiv 0$, S_i is increasing monotonically (even all derivatives are non-negative) as a function of $T_i \geq 0$. If $P_i^+ \equiv 0$, S_i is negative and approaches zero at $T_i \to \infty$. If both P_i^+ and P_i^- have nonzero terms, S_i is negative at small values of T_i, but monotonically increasing, and positive for large values of T_i. Thus, there is a single point at $T_i > 0$ where $S_i = 0$ and F has a minimum value. Then F has a unique smallest value at some point $T_i \geq 0$ ($Y_i^- \leq T_i \leq Y_i^+$). Only in the special case $P_i^+ \equiv P_i^- \equiv 0$ the minimum is not unique (F is independent of T_i, and the economic optimization leads to $T_i = Y_i^+$).

Consider now F as a function of all T_i's. If a minimum point of F at $\mathbf{T}_o$ has been found such that $S_i = 0$ for all i with finite T_{io}, F increases monotonically in every direction from $\mathbf{T}_o$. There can be no other local minima. This proves the uniqueness of the solution, at least in the case of a rare-event approximation.

4.3 Repair costs

In this section we consider how to take into account repair costs associated with any component j belonging to test timing group i.

The probability of failure per one test interval is $\alpha_j + \lambda_j T_i$, where α_j is the failure probability due to a demand (test) and λ_j is the standby failure rate for component j. Thus, the repair cost per unit time is

$$y_{rj} = \frac{\alpha_j}{T_i} b_{\alpha j} + \lambda_j b_{\lambda j}, \tag{39}$$

where $b_{\alpha j}$ and $b_{\lambda j}$ are the mean cost of repair for failures of type α and λ, respectively. Note that the second term is independent of T_i and, therefore, does not have an impact on the optimization of T_i. If the sum $\sum_j y_{rj}$ is added to y in eqn (9) as an additional cost, the form of the optimization problem does not change. The only change needed is to replace C_i with $C_i + \sum_j \alpha_j b_{\alpha j}$, the sum *over components belonging to the timing group i.* The general impact of the additional cost term is to increase the optimum interval T_i. (Thus neglecting the repair cost due to α_j could lead to more frequent nonoptimal testing.)

4.4 Outage costs

Failures of standby safety systems can cause plant outage and loss of production in two ways due to Technical Specifications:

(1) Multiple or common cause failures forcing plant shutdown;

(2) Repair time exceeding the allowed outage time (AOT) of a component.

Assuming that common cause failures cause most of the losses of the first kind, as is usually the case, the cost per unit time is $\sum_k \lambda_k \tau_k C_p$, where the sum is over common cause failures with rates λ_k and repair times τ_k, and C_p is the net value production per unit time. Because this term is independent of any T_i, it does not influence the optimization on the cost function side (y in eqn (9)). Common cause failures are already part of the risk function $F(\mathbf{T})$ and certainly influence optimization, but the cost of outage they cause does not influence optimization.

Partially similar conclusions can be drawn about

losses of the second kind. Usually a small fraction γ_k of component repair times exceeds AOT. Failures during standby can cause outage cost rate $\sum_k \gamma_k \lambda_k \tau_k C_p$, *the sum over all components belonging to the timing group i.* This part is again independent of any T_i and irrelevant for the optimization procedure.

Failures due to a demand (and exceeding AOT) cause a cost rate term

$$\sum_k \frac{\gamma_k \alpha_k \tau_k}{T_i} C_p.$$

When this is included in the cost function, the impact is simply to replace C_i with $C_i + \sum_{k \in K_i} \gamma_k \alpha_k \tau_k C_p$ (increasing the optimal T_i somewhat). The product $\gamma_k \alpha_k$ is often so small that the additional term can be deleted.

5 TOTAL COST OPTIMIZATION

In this section the accident cost is included in the cost function. The first task in Section 5.1 is to find times T_i that yield the absolute minimum total cost rate. Only if the risk at that point exceeds a set criterion R, is it necessary to continue with the constrained optimization described in Section 5.2.

5.1 Unconstrained optimization

The problem is to solve $\mathbf{T} = \mathbf{T}_o$ such that the total cost rate is minimized:

$$y = \sum_{i=1}^{I} \frac{C_i}{T_i} + C_A F(\mathbf{T}) = \min! \qquad (40)$$

where C_A is the cost of an accident.

The functional form of y shows that it is large and decreasing function of any T_i at small values of T_i and increasing for large values of T_i if there is at least one MCS with probability $\sim T_i^n$ with $n > 0$. (If no such term exist, both the risk and the cost are minimized by selecting T_i as large as possible; such T_i can be held constant at Y_i^+ and removed from further optimization.) Thus, there is a minimum of y in the region $T_i > 0$, $i = 1, 2, \ldots, I$.

Equation (40) yields the conditions

$$\frac{C_i}{T_i} - C_A S_i(\mathbf{T}) = 0; \quad i = 1, \ldots, I. \qquad (41)$$

Comparison with eqns (22) shows that the problem is mathematically similar to the risk-limited optimization except that the Lagrange multiplier is now replaced with a known quantity C_A. One can conclude from eqn (41) that (a) every S_i is positive ($i = 1, \ldots, I$) at the optimal point $\mathbf{T}_o$ (because $C_A T_i / C_i > 0$), (b) all cost/risk sensitivity ratios $r_i = C_r/(T_i S_i)$ are equal, to C_A, at the optimal point, (c) only the relative costs C_i/C_A influence the optimization, and (d) the terms of F that are independent of any T_i do not influence the optimization.

Examples:

E.1 With a risk function $F = \sum_{i=1}^{I} a_i T_i^{n_i}$ eqns (41) yield the optimal intervals

$$T_i = \left(\frac{C_i}{n_i a_i C_A}\right)^{1/n_i+1}, \quad i = 1, 2, \ldots, I.$$

The optimal cost rate can be shown to be

$$y = \sum_{i=1}^{I} \frac{n_i + 1}{n_i} \cdot \frac{C_i}{T_i}.$$

E.2 The linear multiple initiating event example (eqn (26)) leads to $T_i = (C_i/a_i C_A)^{1/2}$, and the risk rate at the optimal point is $F = \sum_{i=1}^{I} (a_i C_i / C_A)^{1/2}$. A numerical example with $I = 2$, $a_1 = 1{\cdot}5288 \times 10^{-12}/\text{hr}^2$ (LOCA), $a_2 = 1{\cdot}6987 \times 10^{-13}/\text{hr}^2$ (LOSP), $C_2/C_1 = 9$, $C_A/C_1 = 10^6$ yields $T_1 = 808{\cdot}77$ hr, $T_2 = 7278{\cdot}8$ hr and $F = 2{\cdot}166 \times 10^{-5}/\text{yr}$.

E.3 The product from $F = aT_1T_2$ with eqn (41) yields $T_1 = (C_1^2/aC_2C_A)^{1/3}$, $T_2 = (C_2^2/aC_1C_A)^{1/3}$. The risk and the cost at the optimum point are $F = (aC_1C_2/C_A^2)^{1/3}$ and $y = 3C_AF$, respectively.

E.4 The formalism can be used also for optimizing loss-of-profit risk in alternating production systems, not only accident risks with standby systems. Consider a two-train production system, one train operating and one on standby. The roles of the trains are altered with intervals T. The standby train is maintained with cost C and duration Δ. The rate of production loss events is

$$F = f\left(\rho + \tfrac{1}{2}\lambda T + \frac{\Delta}{T}\right),$$

where f is the failure rate of the operating train, and ρ and λ are parameters for the train on standby. The 'accident' cost can be defined as $C_A = \tau P$, where τ is the average repair time and P is the profit rate for the system in operation. Equation (41) yields optimum T,

$$T_o = \left[2\left(\Delta + \frac{C}{C_A f}\right)\Big/\lambda\right]^{1/2}.$$

(This is the solution also if f is an initiating event rate and there is only one component or train in a safety system, with parameters ρ, Δ, λ).

In all these examples higher values of C_A lead to higher optimal cost, shorter intervals and lower risk.

Analytical solution of eqns (41) is not always feasible. An iterative method can be based on the differential

$$\mathrm{d}y = \sum_{i=1}^{I} \left(-\frac{C_i}{T_i} + C_A S_i\right) \frac{\mathrm{d}T_i}{T_i}. \qquad (42)$$

Effective cost reduction is accomplished by increasing T_i when the coefficient $-C_i/T_i + C_A S_i$ is most negative, or decreasing T_i when the coefficient is most positive. Throughout this process one has to observe the limits $Y_i^- \leq T_i \leq Y_i^+$, of cause. In the end $r_i \cong C_A$ for all i such that $Y_i^- < T_i < Y_i^+$.

After the minimum of y has been found, one has to verify that $F(\mathbf{T}_o) \leq R$. If $F(\mathbf{T}_o) > R$, one can try changing any consecutive testing schemes to staggered testing, because this usually reduces risk, and try optimization again. If still $F(\mathbf{T}_o) > R$, one has to find a new point $\mathbf{T}$ where y is the smallest under the constraint $F \leq R$. Figure 2 illustrates the general behaviour of the cost and risk functions in terms of a particular T_i. In a normal case $F(\mathbf{T}_o) \leq R$, and the solution is found.

If $F(\mathbf{T}_o) > R$, one has to reduce T_i until the risk limit is satisfied. This is explained in the next section.

5.2 Risk-constrained opimization

If the absolute minimization of y yields a risk rate that is larger than R, eqn (40) has to be replaced with

$$x = \sum_{i=1}^{I} \frac{C_i}{T_i} + C_A F + \hat{L}(F - R) = \min! \qquad (43)$$

where $\hat{L} > 0$ is a Lagrange multiplier. Equation (41) is then replaced by

$$\frac{C_i}{T_i} - (C_A + \hat{L})S_i = 0, \quad i = 1, 2, \ldots, I. \qquad (44)$$

At a point where eqn (44) is satisfied, $dy = -\hat{L}\, dF$. Thus, it is not possible to reduce the cost rate y any further by reducing F below R.

When eqn (44) is compared with the earlier optimization eqn (22), one observes that definition $L = \hat{L} + C_A$ creates a complete equivalence between the two. This yields the interpretation that L in eqn (22) is the 'effective' accident cost. Thus, the optimization procedure similar to Appendix 1 can be applied (starting now from step 5). It leads to $r = C_A + \hat{L}$ and to the minimum of $y = \sum_{i=1}^{I} C_i/T_i + C_A F$ if the minimum of F is not larger than R. If the minimum of F is larger than R, the process ends up with $S_i \cong 0$ for all i ($r_i \to \infty$), in which case there is no acceptable solution.

Analytical solutions for several simple yet typical risk functions are given in Table 1.

In case of example E.2 of Section 5.1 one can conclude that the risk limit $R = 10^{-5}$/yr can be satisfied if C_A is replaced with $C_A + \hat{L}$ which is $(2{\cdot}166)^2 = 4{\cdot}69$ times the original C_A. The times T_1 and T_2 are reduced to 373 hr and 3360 hr, respectively, and the total cost is increased by a factor of 2·166.

In all examples E.1–E.3 risk can be reduced to any value $R > 0$ by controlling the value of $C_A + \hat{L}$ in place of C_A. Thus, one can call $\hat{L}$ the 'virtual' accident cost and $C_A + \hat{L}$ the 'effective' total accident cost dictated by a risk criterion, R. The maintenance example of Appendix 2 yielded cost/risk sensitivity ratios $r_i \cong r = 137 \times 10^6$ \$, which is the effective accident cost dictated by $R = 10^{-4}$/yr.

In example E.4 there is a limit to which F can be reduced by increasing C_A. The limit is the absolute minimum of F.

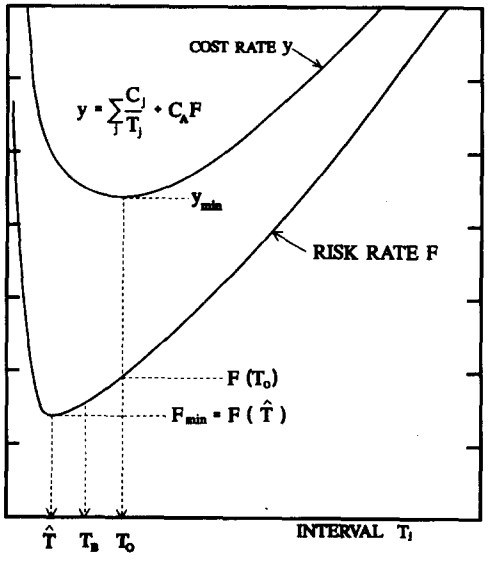


Fig. 2. Optimum intervals T_i: (1) T_o is the solution when $R \geq F(T_o)$, (2) T_B is the solution, $F(T_B) = R$, when $F_{min} \leq R \leq F(T_o)$, (3) there is no solution if $R < F_{min}$.

6 CONCLUSION

A systematic procedure has been developed for optimizing test and maintenance intervals by minimizing the total testing and maintenance costs while satisfying a risk criterion. The measure of risk is the time-average accident frequency. Component failures, common cause failures and human errors are included and modeled by basic events, the probabilities of which are simple functions of test and maintenance intervals. The procedure is formulated in terms of minimum cut sets (or terms) and basic events, subjects that are easily manipulated by computer codes solving fault trees. Analytical solutions have been obtained for several special risk models, illustrating how different factors influence the optimization. Essential features of the methodology include combined consideration of risk and cost, inclusion of maintenance effects in addition to test periods, and versatility in handling operating and standby components and various failures modes. The impacts of repair costs and outage times have also been considered. Examples have shown that significant economical benefits can be obtained by optimizing test and maintenance intervals. It has been shown that

Table 1. Optimal time intervals T_i, minimum costs and Lagrange multipliers (effective accident costs)

Risk function	Cost function	
$F(\mathbf{T})$	$y = \sum_{i=1}^{I} \frac{C_i}{T_i} + C_A F = \min!$	$y = \sum_{i=1}^{I} \frac{C_i}{T_i} = \min!$, with $F \le R$
aT_1T_2	$T_1 = [C_1^2/(aC_2C_A)]^{1/3}$ $T_2 = [C_2^2/(aC_1C_A)]^{1/3}$ $y = 3C_AF$	$T_1 = [RC_1/(aC_2)]^{1/2}$ $T_2 = [RC_2/(aC_1)]^{1/2}$ $L = (aC_1C_2)^{1/2}/R^{3/2}$
$\sum_{i=1}^{I} a_iT_i$	$T_i = \left(\frac{C_i}{a_iC_A}\right)^{1/2}$	$L = \left(\sum_{j=1}^{I} \sqrt{a_jC_j}\right)^2 \Big/ R^2$
	$y = 2C_AF$	$T_i = \left(\frac{C_i}{a_iL}\right)^{1/2}$
$\sum_{i=1}^{I} a_iT_i^{n_i}$	$T_i = \left(\frac{C_i}{n_ia_iC_A}\right)^{1/n_i+1}$, $y = \sum_{i=1}^{I} \frac{n_i+1}{n_i} \cdot \frac{C_i}{T_i}$	$T_i = \left(\frac{C_i}{n_ia_iL}\right)^{1/n_i+1}$, where L is a solution of $\sum_{i=1}^{I} a_i\left(\frac{C_i}{n_ia_iL}\right)^{n_i/n_i+1} = R$
$f\left(\rho + \frac{1}{2}\lambda T + \frac{\Delta}{T}\right)$	$T = \left[2\left(\Delta + \frac{C}{fC_A}\right)\Big/\lambda\right]^{1/2}$, when $R \ge F(T)$. If $R < F(T)$, use next column.	$T = (d + \sqrt{d^2-1})(2\Delta/\lambda)^{1/2}$ when $d = \frac{R - f\rho}{f\sqrt{2\lambda\Delta}} \ge 1$, $L = \frac{C}{2f\Delta(d^2 - 1 + d\sqrt{d^2-1})}$
		if $d < 1$, there is no solution $[R < F(\hat{T})]$.

realistic optimization of maintenance is possible if the risk model includes beneficial effects of maintenance in addition to any unavailability caused by maintenance. Including the cost of an accident in the formalism does not essentially complicate the problem. In many cases it can make the solution analytically easier. In summary, the following general conclusions can be drawn from this study:

(1) The optimal test and maintenance intervals depend only on the relative cost values rather than the absolute values. This makes a solution rather insensitive to systematic errors in the cost estimates.
(2) The intervals based on minimum cost are generally larger than those based on minimum risk.
(3) Overestimating the cost of an individual test or maintenance action leads to an extended interval for that particular action (while reducing the intervals for other actions, in some cases). However, a specified risk limit always guarantees that unacceptably long intervals are avoided.
(4) The minima of the risk and cost functions are often rather flat. Therefore the consequences of small errors are not severe. Furthermore, sensitivities to small errors can be easily evaluated in case analytical solutions are available (e.g., Table 1).
(5) Increasing the accident cost estimate leads to a higher total cost, shorter intervals and a lower accident rate. Thus, the accident cost parameter can be used to control the risk.
(6) The Lagrange multiplier of an optimization problem has several interesting interpretations: it is the effective accident cost as well as a common cost/risk sensitivity ratio (in case the accident cost has not been specified in advance, or if minimizing the total cost leads to a risk higher than the limit).

It is possible that the owner of a plant and the authorities ('society') have different ideas about the cost of an accident. Thus, they can come to different conclusions about acceptable test and maintenance intervals. Specifying a risk limit instead of an accident cost allows proper freedom for the interval optimization.

REFERENCES

1. Jacobs, I. M., Reliability of engineered safety features as a function of testing frequency. *Nuclear Safety*, **9** (1968) 303–312.
2. Hirsch, H. M., Setting test intervals and allowable bypass times as a function of protection system goals. *IEEE Trans. Nuclear Science*, **N-18** (1971) 488–494.
3. Signoret, J. P., Availability of a periodically tested standby system, *NUREG/TR-0027* 1976.

4. Vaurio, J. K., Comments on system availability analysis and optimal test intervals. *Nucl. Engng Design,* **128** (1991) 401–402.
5. Vaurio, J. K. & Sciaudone, D., Unavailability modeling and analyis of redundant safety systems. *ANL-79-87,* Argonne National Laboratory 1979.
6. Vesely, W. E. & Goldberg, F. F., FRANTIC—A computer code for time dependent unavailability analysis. *NUREG-0193* 1977.
7. IAEA-TECDOC-737, Advances in reliability analysis and probabilistic safety assessment for nuclear power reactors. Technical Committee Report, Budapest, 7–11 September 1992, International Atomic Energy Agency (1994).
8. Kim, I. S., Martorell, S., Vesely, W. E. & Samanta, P. K., Quantitative evaluation of surveillance test intervals including test-caused risks. *NUREG/CR-5775, BNL-NUREG-52296* 1992.
9. Čepin, M., Košuh, M. & Mavko, B., Risk impacts associated with surveillance tests. In *Proc. 4th TÜV-Workshop on Living PSA Application,* Hamburg, 2–3 May 1994, Technischer Überwachungs-Verein Nord e.V. 1994.
10. Uryašev, S., On optimization of test strategies. In *Proc. PSA'93,* Clearwater Beach, January 26–29, 1993, American Nuclear Society 1993.
11. Sandstedt, J., Demonstration case studies on living PSA. In *Proc. PSA'93,* Clearwater Beach, January 26–29, 1993, American Nuclear Society 1993.
12. Robertson, M. & Byrne, J., Converting a static PSA for the Dounreay prototype fast reactor into a living PSA. In *Proc. PSA/PRA and Severe Accidents '94,* 17–20 April 1994, Ljubljana, Nuclear Society of Slovenia 1994.
13. Levinson, S. H. & Enzinna, R. S., ESFAS reliability analysis to justify test interval extension. In *Proc. PSAM—II,* San Diego, March 20–25, 1994, Society for Risk Analysis (1994).
14. Papazoglou, I. A., Bari, R. A., Buslik, A. J., Hall, R. E., Ilberg, D., Samanta, P. K., Teichmann, T., Youngblood, R. W., El-Bassioni, A., Fragola, J., Lofgren, E. & Vesely, W., Probabilistic safety analysis procedures guide. *NUREG/CR-2815,* Section 5.6.3, U.S. Nuclear Regulatory Commissions (1984).
15. Vaurio, J. K., Developments in reliability data collection and analysis. In *Proc. IFAC Symp. SAFEPROCESS 94,* Espoo, Finland, 13–15 June, 1994.
16. Vaurio, J. K., The theory and quantification of common cause shock events for redundant standby systems. *Reliab. Engng & System Safety,* **43** (1994) 289–305.
17. Apostolakis, G. E. & Bansal, P. P., Effect of human error on the availability of periodically inspected redundant systems. *IEEE Trans. Reliab.* **R-26** (1977) 220–225.
18. Swain, A. D. & Guttmann, H. E., Handbook of human reliability analysis with emphasis on nuclear power plant applications. *NUREG/CR-1278,* U.S. Nuclear Regulatory Commission 1983.
19. Vesely, W. E., Quantifying maintenance effects on unavailability and risk using Markov Modeling. *Reliab. Engng & System Safety,* **41** (1993) 177–187.

APPENDIX 1: ITERATIVE PROCEDURE FOR SOLVING OPTIMAL INTERVALS

Setting up the problem:

A. Select I^* timing groups i ($i = 1, 2, \ldots, I^*$) for which the times T_i are subject to optimization (i.e. not fixed based on considerations other than optimization). Determine also the limits of variation $Y_i^- \le T_i \le Y_i^+$, $i = 1, 2, \ldots, I^*$. Values within this interval are called admissible. (For example, Y_i^+ can be 1 year for testing, 4 years for major maintenance overhaul, etc.).

B. Select risk criterion R.
Special case: calculate risk at $\mathbf{T} = \mathbf{Y}^+$ or otherwise verify that $F(\mathbf{Y}^+) > R$. If $F(\mathbf{Y}^+) \le R$, the optimal solution is $\mathbf{T} = \mathbf{Y}^+$.

C. Identify basic events (indices k) that belong to each subset K_i^+ (i.e. the unavailability $u_k \sim T_i$), and to each $K_i^-(u_k \sim T_i^{-1})$, for all timing groups i ($i = 1, 2, \ldots, I^*$).

D. If there is no basic event with probability proportional to a particular T_j (K_j^+ is empty), set this T_j to the maximum value ($T_j = Y_j^+$) permanently, and eliminate T_j from further consideration. Calculate new $F(\mathbf{T})$ if any T_j was changed. (This step does not increase risk or cost but can reduce both because $S_j \le 0$ in the whole admissible range no matter what values $T_i > 0$ the other variables have. The number of remaining timing groups is $I \le I^*$. Determine the cost parameters C_i for $i = 1, 2, \ldots, I$.

E. It is permissible, but not necessary, to delete from the risk rate F any terms that are independent of all free variables T_j. R has to be reduced by the same amount. (e.g. terms that do not contain any basic events in any K_i^+ or K_i^- can be subtracted.)

Optimization procedure

1. Calculate all $S_i(\mathbf{T})$, $i = 1, 2, \ldots, I$ (eqn (20)). If any $S_i \le 0$, increase the corresponding T_i until $S_i > 0$ or until $T_j = Y_i^+$.
2. Determine $F = F(\mathbf{T})$, cost rates C_i/T_i and $y(\mathbf{T})$ (eqn (6)), sensitivities $S_i(\mathbf{T})$ (eqn (20)), cost/risk sensitivity ratios r_i (eqn (31)) and r (eqn (32)). If any $S_i < 0$ while $T_i < Y_i^+$, repeat from step 1 until no T_i is changed. At the end of this step $T_j = Y_j^+$ for any j such that $S_j(\mathbf{T}) \le 0$.
3. If $F \cong R$, go to step 6.
4. If $F < R$ (while $F(\mathbf{Y}^+) > R$): Identify *imax* such that r_{imax} = largest postiive r_i for which $T_i < Y_i^+$ ($i = 1, 2, \ldots, I$). (There is at least one positive r_i unless all $S_i \le 0$ and $\mathbf{T} = \mathbf{Y}^+$ is the solution.) Increase T_{imax} a reasonable increment ΔT_{imax} such that

$$0 < \frac{\Delta T_{imax}}{T_{imax}} \le \min\left\{\frac{R - F}{S_{imax}}, \frac{Y_i^+ - T_{imax}}{T_{imax}}, \frac{1}{2}\right\}. \tag{A1.1}$$

This assures that F approaches R without T_{imax} exceeding Y_{imax}^+.
Go to step 2.
5. If $F > R$:
Verify: If all T_i's are either $= Y_i^-$ or Y_i^+, or if the ratios S_i/F for all positive S_i's ($T_i < Y_i^+$) are small,

e.g. <0·01, then the risk cannot be essentially reduced (see eqn (29)), and solution does not exist. Otherwise:

Identify *imin* such that that r_{imin} = smallest positive r_i for which $T_i > Y_i^-$ ($i = 1, 2, \ldots, I$). Reduce T_{imin}. The absolute value of a change should be limited,

$$0 < -\frac{\Delta T_{imin}}{T_{imin}} \leq \min\left\{\frac{F-R}{S_{imin}}, \frac{T_{imin}-Y_{imin}^-}{T_{imin}}, \frac{1}{2}\right\}. \quad \text{(A1.2)}$$

Go to step 2.

6. At this point one has $F \cong R$ (unless one has exited the procedure because $\mathbf{T} = \mathbf{Y}^+$ or because the minimum of F has been reached with $R < F$). If there are less than two values of T_i satisfying $Y_i^- < T_i < Y_i^+$, no further optimization is possible: the optimum vector $\mathbf{T}$ has been found. If there are at least two values T_i with $Y_i^- < T_i < Y_i^+$ (with corresponding $S_i > 0$), continue from step 7.
7. Definite indices *imax* and *imin* such that r_{imax} = largest positive r_i for which $T_i < Y_i^+$, r_{imin} = smallest positive r_i for which $T_i > Y_i^-$, $i = 1, 2, \ldots, I$.
8. If $(r_{imax} - r_{imin})/r$ is small, e.g. < 0·02, one has reached the optimal solution $\mathbf{T}$ and the optimal cost $y = \sum_{i=1}^{I} C_i/T_i$. Otherwise, go to step 9.
9. The idea now is to increase T_{imax} (to reduce cost most effectively) and reduce T_{imin} (to increase cost least effectively) so that the risk does not change as a net result, $\Delta F \approx 0$. This can be accomplished by selecting changes ΔT_{imax} and ΔT_{imin} that satisfy eqn (33). A practical suggestion is to calculate new values

$$T_{imax,new} = \min\left\{\sqrt{\frac{r_{imax}}{r}}\, T_{imax}, Y_i^+\right\} \quad \text{(A1.3)}$$

$$T_{imin,new} = \max\left\{T_{imin}\left[1 - \frac{S_{imax}}{S_{imin}}\left(\frac{T_{imax,new}}{T_{imax}} - 1\right)\right], Y_i^-\right\} \quad \text{(A1.4)}$$

Go to step 2.

Note for a special case: If there are no basic events with probabilities inversely proportional to any T_i (all K_i^- are empty), the maximum risk rate is $F(\mathbf{Y}^+)$ and the minimum is $F(\mathbf{Y}^-)$. If $R < F(\mathbf{Y}^-)$, there is no solution. If $R \geq F(\mathbf{Y}^+)$, then $\mathbf{T} = \mathbf{Y}^+$ is the solution. In the range $F(\mathbf{Y}^-) \leq R < F(\mathbf{Y}^+)$ a solution exists and $S_i \geq 0$ for all $i = 1, 2, \ldots, I$ in the admissible region.

APPENDIX 2: STEP BY STEP SOLUTION OF THE MAINTENANCE PROBLEM

A. $Y_1^- = 0$, $Y_1^+ = 2000$ hr, $Y_2^- = 2000$ hr, $Y_2^+ = 20\,000$ hr.
B. $F(\mathbf{Y}^+) = 3{\cdot}1 \times 10^{-4}$/yr $> R$.
C. $K_1^+ = \{1\}$, $K_1^- = \{\ \}$ (empty), $K_2^+ = \{2\}$, $K_2^- = \{3\}$.
D. $C_1 = 1000$ \$, $C_2 = 4000$ \$.
1. $S_1^+ = fu_1 + fu_1u_2$, $S_1^- = 0$, $S_2^+ = fu_1u_2$, $S_2^- = fu_3$, $S_1 = fu_1 + fu_1u_2 = f\frac{1}{2}\lambda_o T_1(1 + \varepsilon T_2)$

$$S_2 = fu_1u_2 - fu_3 = f\left(\tfrac{1}{2}\lambda_o T_1 \varepsilon T_2 - \frac{\Delta}{T_2}\right).$$

Initial values: $T_1 = 2000$ hr, $T_2 = 20\,000$ hr ($\mathbf{T} = \mathbf{Y}^+$).

2. $F = 3{\cdot}1 \times 10^{-4}$/yr, $S_1 = 3 \times 10^{-4}$/yr (>0), $S_2 = 1{\cdot}9 \times 10^{-4}$/yr (>0), y = 0·7 \$/hr, $r_1 = 1666{\cdot}7$ \$. yr/hr, $r_2 = 1052{\cdot}6$ \$. yr/hr.
5. $F > R$, $r_2 < r_1 \rightarrow imin = 2$, $-\Delta T_2/T_2 = \frac{1}{2}$, $T_2 = 10\,000$ ($T_1 = 2000$).
2. $F = 2{\cdot}2 \times 10^{-4}$, $S_1 = 2 \times 10^{-4}$ (>0), $S_2 = 0{\cdot}8 \times 10^{-4}$ (>0), $y = 0{\cdot}9$, $r_1 = 2500{\cdot}0$, $r_2 = 5000{\cdot}0$.
5. $r_1 < r_2 \rightarrow imin = 1$, $-\Delta T_1/T_1 = \frac{1}{2} \rightarrow T_1 = 1000$ ($T_2 = 10\,000$).
2. $F = 1{\cdot}2 \times 10^{-4}$, $S_1 = 1 \times 10^{-4}$ (>0), $S_2 = 0{\cdot}3 \times 10^{-4}$ (>0), $y = 1{\cdot}4$, $r_1 = 10\,000$, $r_2 = 13\,333$.
5. $r_1 < r_2 \rightarrow imin = 1$, $-\Delta T_1/T_1 = 0{\cdot}2 \rightarrow T_1 = 800$ ($T_2 = 10\,000$).
2. $F = 1 \times 10^{-4}$, $S_1 = 0{\cdot}8 \times 10^{-4}$ (>0), $S_2 = 0{\cdot}2 \times 10^{-4}$ (>0), $y = 1{\cdot}65$, $r_1 = 15\,625$, $r_2 = 20\,000$, $r = 16\,500$.
3. $\rightarrow 6. \rightarrow 7.$ $r_1 < r_2 \rightarrow imax = 2$, $imin = 1$.
8. $(r_2 - r_1)/r = 0{\cdot}265$ (not small).
9. New $T_2 = 12\,121$, new $T_1 = 757{\cdot}58$.
2. $F = 1{\cdot}0029 \times 10^{-4}$, $S_1 = 0{\cdot}83792 \times 10^{-4}$, $S_2 = 0{\cdot}29413 \times 10^{-4}$, $y = 1{\cdot}65$, $r_1 = 15\,753$, $r_2 = 11\,219$, $r = 14\,575$.
5. $r_2 < r_1 < imin = 2$, $-\Delta T_2/T_2 = 0{\cdot}00994 \rightarrow T_2 = 12\,000$ ($T_1 = 757{\cdot}58$).
2. $F = 1{\cdot}0000 \times 10^{-4}$, $S_1 = 0{\cdot}8333 \times 10^{-4}$, $S_2 = 0{\cdot}2879 \times 10^{-4}$, $y = 1{\cdot}6533$, $r_1 = 15\,840$, $r_2 = 11\,579$, $r = 14\,746$.
3. $\rightarrow 6. \rightarrow 7.$ $r_2 < r_1 \rightarrow imax = 1$, $imin = 2$.
8. $(r_1 - r_2)/r = 0{\cdot}289$ (not small).
9. New $T_1 = 785{\cdot}18$, new $T_2 = 10\,734$.
2. $F = 1{\cdot}000 \times 10^{-4}$, $S_1 = 0{\cdot}8140 \times 10^{-4}$, $S_2 = 0{\cdot}2351 \times 10^{-4}$, $y = 1{\cdot}646$, $r_1 = 15\,646$, $r_2 = 15\,850$, $r = 15\,692$.
3. $\rightarrow 6. \rightarrow 7.$ $r_1 < r_2 \rightarrow imax = 2$, $imin = 1$.
8. $(r_2 - r_1)/r = 0{\cdot}013$ (small).